Cambridge Assessment International Education

# Cambridge International AS & A Level

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*3967186601*

**MATHEMATICS** **9709/11**

Paper 1 Pure Mathematics 1 **May/June 2023**

**1 hour 50 minutes**

You must answer on the question paper.

You will need: List of formulae (MF19)

**INSTRUCTIONS**

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

**INFORMATION**

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **20** pages. Any blank pages are indicated.


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**1** Solve the equation $4 \sin \theta + \tan \theta = 0$ for $0° < \theta < 180°$. [3]

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**2** **(a)** Find the first three terms in the expansion, in ascending powers of $x$, of $(2 + 3x)^4$. [2]

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**(b)** Find the first three terms in the expansion, in ascending powers of $x$, of $(1 - 2x)^5$. [2]

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**(c)** Hence find the coefficient of $x^2$ in the expansion of $(2 + 3x)^4(1 - 2x)^5$. [2]

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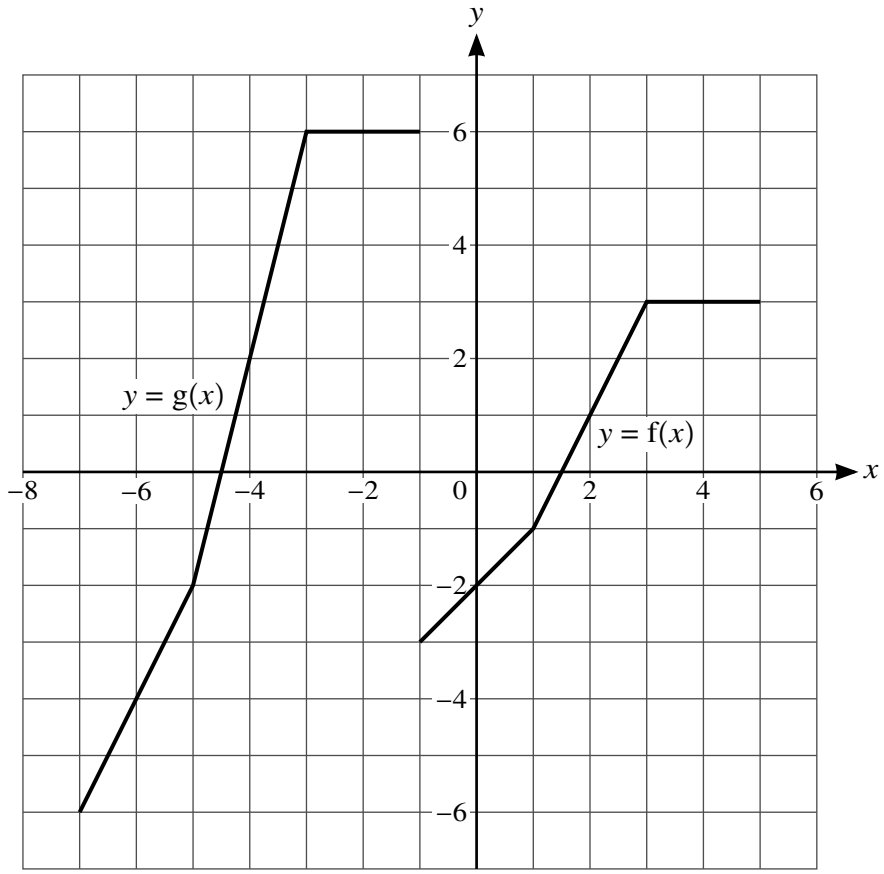
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**3**




The diagram shows graphs with equations $y = f(x)$ and $y = g(x)$.

Describe fully a sequence of two transformations which transforms the graph of $y = f(x)$ to $y = g(x)$. [4]

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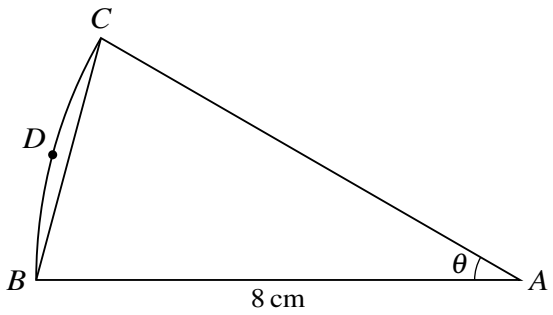
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**4**




The diagram shows a sector $ABC$ of a circle with centre $A$ and radius 8 cm. The area of the sector is $\frac{16}{3}\pi\,\text{cm}^2$. The point $D$ lies on the arc $BC$.

Find the perimeter of the segment $BCD$. [4]

**5** The line with equation $y = kx - k$, where $k$ is a positive constant, is a tangent to the curve with equation $y = -\frac{1}{2x}$.

Find, in either order, the value of $k$ and the coordinates of the point where the tangent meets the curve. [5]

**6** The first three terms of an arithmetic progression are $\frac{p^2}{6}$, $2p - 6$ and $p$.

**(a)** Given that the common difference of the progression is not zero, find the value of $p$. [3]

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**(b)** Using this value, find the sum to infinity of the geometric progression with first two terms $\frac{p^2}{6}$ and $2p - 6$. [2]

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**7** A curve has equation $y = 2 + 3\sin\frac{1}{2}x$ for $0 \leqslant x \leqslant 4\pi$.

**(a)** State greatest and least values of $y$. [2]

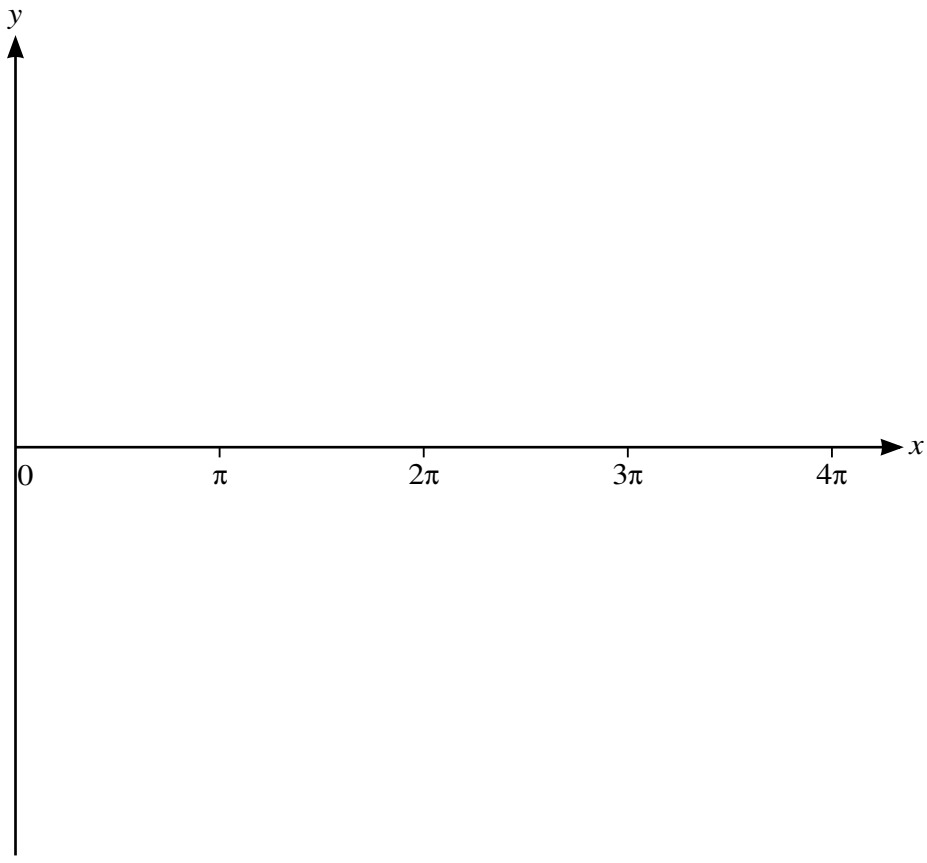
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**(b)** Sketch the curve. [2]




**(c)** State the number of solutions of the equation

$$2 + 3\sin\tfrac{1}{2}x = 5 - 2x$$

for $0 \leqslant x \leqslant 4\pi$. [1]

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**8** The functions f and g are defined as follows, where $a$ and $b$ are constants.

$$\mathrm{f}(x) = 1 + \frac{2a}{x - a} \text{ for } x > a$$

$$\mathrm{g}(x) = bx - 2 \text{ for } x \in \mathbb{R}$$

**(a)** Given that $\mathrm{f}(7) = \frac{5}{2}$ and $\mathrm{gf}(5) = 4$, find the values of $a$ and $b$. [4]

For the rest of this question, you should use the value of $a$ which you found in **(a)**.

**(b)** Find the domain of $f^{-1}$. [1]

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**(c)** Find an expression for $f^{-1}(x)$. [3]

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**9** Water is poured into a tank at a constant rate of $500\,\text{cm}^3$ per second. The depth of water in the tank, $t$ seconds after filling starts, is $h$ cm. When the depth of water in the tank is $h$ cm, the volume, $V\,\text{cm}^3$, of water in the tank is given by the formula $V = \frac{4}{3}(25 + h)^3 - \frac{62\,500}{3}$.

**(a)** Find the rate at which $h$ is increasing at the instant when $h = 10$ cm. [3]

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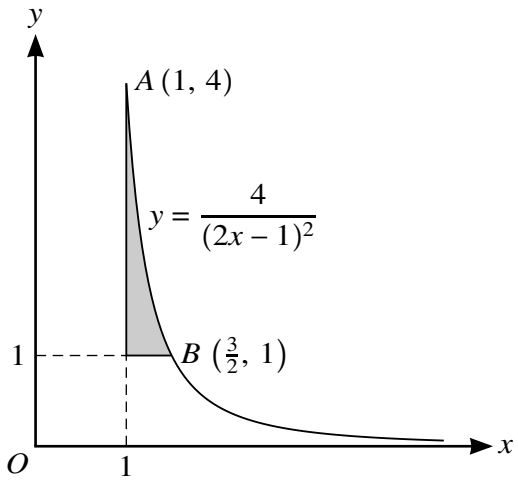
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**(b)** At another instant, the rate at which $h$ is increasing is 0.075 cm per second.

Find the value of $V$ at this instant. [3]

**10**




The diagram shows part of the curve with equation $y = \dfrac{4}{(2x-1)^2}$ and parts of the lines $x = 1$ and $y = 1$. The curve passes through the points $A\,(1,\ 4)$ and $B, \left(\frac{3}{2},\ 1\right)$.

**(a)** Find the exact volume generated when the shaded region is rotated through 360° about the $x$-axis. [5]

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**(b)** A triangle is formed from the tangent to the curve at $B$, the normal to the curve at $B$ and the $x$-axis.

Find the area of this triangle. [6]

**11** The equation of a curve is such that $\frac{dy}{dx} = 6x^2 - 30x + 6a$, where $a$ is a positive constant. The curve has a stationary point at $(a, -15)$.

**(a)** Find the value of $a$. [2]

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**(b)** Determine the nature of this stationary point. [2]

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**(c)** Find the equation of the curve. [3]

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**(d)** Find the coordinates of any other stationary points on the curve. [2]

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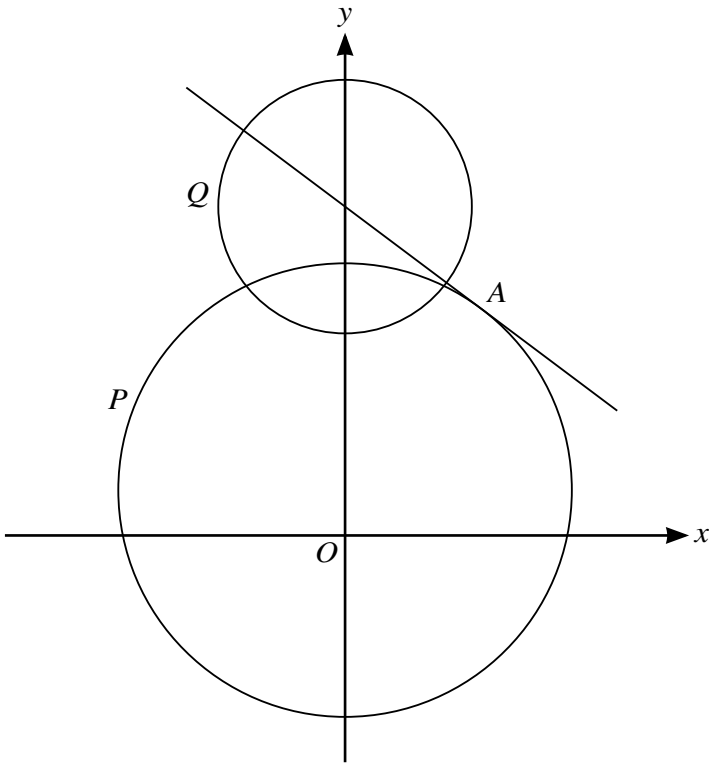
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**12**




The diagram shows a circle $P$ with centre (0, 2) and radius 10 and the tangent to the circle at the point $A$ with coordinates (6, 10). It also shows a second circle $Q$ with centre at the point where this tangent meets the $y$-axis and with radius $\frac{5}{2}\sqrt{5}$.

**(a)** Write down the equation of circle $P$. [1]

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**(b)** Find the equation of the tangent to the circle $P$ at $A$. [2]

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**(c)** Find the equation of circle $Q$ and hence verify that the $y$-coordinates of both of the points of intersection of the two circles are 11. [3]

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**(d)** Find the coordinates of the points of intersection of the tangent and circle $Q$, giving the answers in surd form. [3]

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## Additional Page

If you use the following lined page to complete the answer(s) to any question(s), the question number(s) must be clearly shown.

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